Full Stack Development with MERN

Project Documentation format

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1. Introduction

Project Title: LernHub

Team Members:
Aparna – Frontend Developer

Siri – Data Integration
Deepika – Backend Developer
Lalitha – Database Management

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2. Project Overview

Purpose:
LernHub is developed to help students and self-learners access structured learning resources in one place. The main goal is to provide a centralized platform where users can enroll in courses, follow a learning path, and track their progress efficiently.

Features:
User registration and login
Dashboard to view enrolled courses
Course browsing and enrollment
Progress tracking system
Profile management
Basic course recommendations

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3. Architecture

Frontend:
The frontend is developed using React.js. It provides a responsive user interface with components such as login, registration, dashboard, and course pages. React handles routing and state management for a smooth user experience.

Backend:
The backend is built using Node.js and Express.js. It handles API requests, user authentication, course management, and communication with the database.

Database:
MongoDB is used as the database. It stores user details, course data, and progress information. Collections include Users, Courses, and Enrollments.

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4. Setup Instructions

Prerequisites:
Node.js
MongoDB
VS Code

Installation:
Step 1: Clone the repository
git clone https://github.com/your-repo/lernhub
Step 2: Navigate to project folder
cd lernhub
Step 3: Install backend dependencies
cd server
npm install
Step 4: Install frontend dependencies
cd client
npm install
Step 5: Configure environment variables
Create a .env file in server folder and add MongoDB URI and JWT secret

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5. Folder Structure

Client:
Contains React frontend code including components, pages, and styles

Server:
Contains backend code including routes, controllers, models, and configuration

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6. Running the Application

Frontend:
cd client
npm start

Backend:
cd server
npm start

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7. API Documentation

POST /register
Registers a new user

POST /login
Authenticates user and returns token

GET /courses
Fetches all available courses

POST /enroll
Enrolls user in a course

GET /progress
Fetches user progress

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8. Authentication

Authentication is handled using JSON Web Tokens (JWT).
Users log in with email and password, and a token is generated.
This token is used to access protected routes and maintain user sessions securely.

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9. User Interface

The application includes the following UI screens:
Login page
Registration page
Dashboard
Course listing page
Profile page

Screenshots can be added here after UI development.

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10. Testing

Testing is performed using manual testing methods.
Functional testing is done to verify features like login, registration, and course enrollment.
User Acceptance Testing (UAT) is conducted to ensure the system meets user requirements.

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11. Screenshots or Demo

Screenshots of the application interfaces should be added here.
Alternatively, a demo video link can be provided.

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12. Known Issues

Minor delay in loading dashboard
Limited course data
No advanced recommendation system

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13. Future Enhancements

Add AI-based course recommendations
Implement mobile application
Add quizzes and certification system
Improve UI/UX design
Add real-time notifications